CS & IT ENGINEERING



Basics Of Computer Computer System

Components of Computer

Lecture No.- 01



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Topics to be Covered



- **Topic** Computer Architecture
- **Topic** Computer Organization
- **Topic** Components of Computer
- **Topic** CPU, Memory & IO Devices



Topic : Introduction

- **GATE Ranks:**
  - 682 (2009) – 3rd year
  - 19 (2010) – 4th year
  - 119, 440 etc.

- **Education:**
  - ME from IISc Bangalore
  - M. tech from BITS-pilani in Data Science

- **Work:**
  - 17+ Year Teaching Experience
  - 13+ in GATE/IES
  - Worked in Cisco, Audience Communication



Topic : Syllabus

Lecture Number	Topic
1	Components of Computer
2	How a Computer Works
3	Data in Computer System
4	Number System
5	Binary and Power of 2s
6	Units & Conversions
7	Various Terminologies
8	Memory System
9	Basics of Operating System
10	Basics of Data Structures



Topic : Computer Organization & Architecture

Computer Architecture:

- Conceptual design and fundamental operational structure.

↓

what components
&
what each component does



Topic : Computer Organization & Architecture

Computer Architecture:

- Conceptual design and fundamental operational structure.

Implementation

Computer Organization:

- Deals with physical devices and their interconnections
- With a perspective of improving the performance.



Topic : Components of Computer → Basic

- CPU (Central Processing Unit) :- (master of Computer)

  (Processor) or (Microprocessor)

  → CU (Control Unit)
  → ALU (Arithmetic Logic Unit)

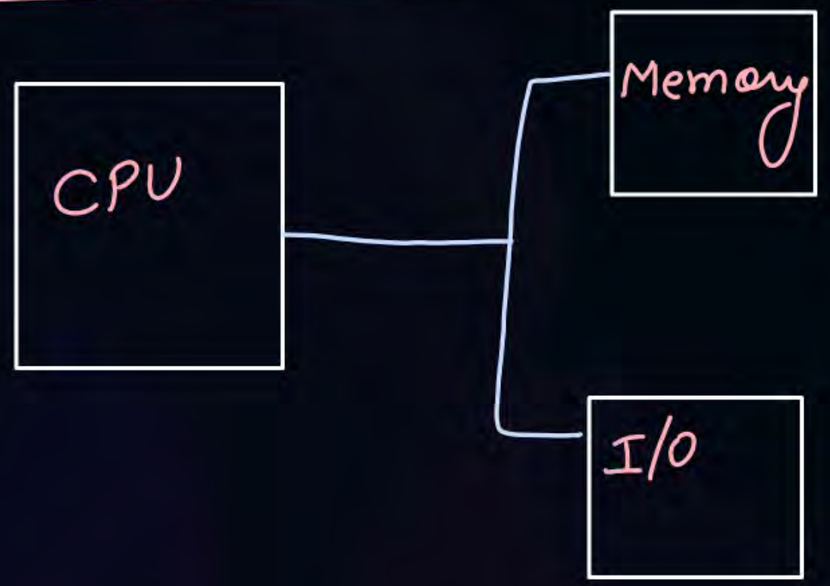
- Memory:
  → Primary (main) memory ⇒ RAM (Temporary Memory)
  → Secondary (Auxillary memory) ⇒ Hard disk, SSD (Permanent Memory)

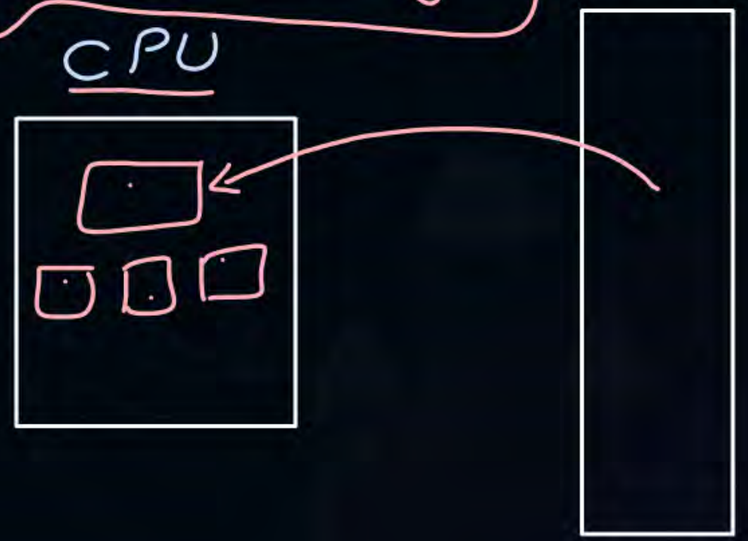
- I/O Devices: (Input-output)
  → Input devices :- keyboard, mouse, camera, microphone
  → output devices :- monitor, printer, Speaker
  → storage devices :- pendrive, CD/DVD



Topic : Other Components

- System Buses ⇒ wired connection between components of computer
  bus contains more than one wire (line).
- CPU Registers
  ↳ small memories inside CPU to carry out processing.







2 mins Summary

Topic Computer Architecture

Topic Computer Organization

Topic Components of Computer

Topic CPU, Memory & IO Devices

Happy Learning

THANK - YOU